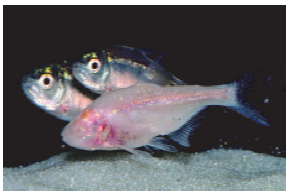
Chapter-17

Habitat



17.1 Meaning of Habitat

We have Kotumsar caves near the banks of Kanker river at Jagdalpur in our state. Long thin conical structures hanging from the roofs and rising from the floors of the caves are formed as lime water trickles down (See Fig.1).

These caves are over a km in length and are located about 35 meters below the surrounding areas. A hand torch carried down into the cave if switched off would result in pitch darkness in the cave.

The amount of oxygen inside the cave is less compared to that outside it. In spite of such adverse conditions, some organisms like bats, a type of fish, insects, millipedes, snails, bacteria, are found etc. in the cave.



Fig. 1: conical structures, the stalactites and stalagmites of Kotumsar caves.

The temperature inside the cave ranges between 25°C- 32°C throughout the year. The temperature of water inside the cave ranges between 22°C - 30°C. This implies that the temperature of water and air inside the cave is similar. During rainy season, the cave gets flooded.

A particular species of fish is found in these caves. The length of the fish is 2-4cm. Its eyes are very small. The local people call it "kaani machri". Other varieties of this fish are found in mountainous streams. It feeds on microscopic organisms, insects and molluscs. It also feeds on dead and decaying remains of animals and plants. Thus it may be called as a scavenger. These fishes usually live in burrows under the soil of the caves. Though their gaseous exchange occurs in water (with the help of gills) as most other fishes, they often come to the surface of water to fill their mouths with air(since they also have air/swim bladders). They lead their whole life in these caves. They reproduce here and die here. Thus we may say that water of the Kotumsar caves is the habitat of these fishes.



Fig. 2 : kaani machri

- What special features of Kotumsar caves have been described in this section?
- Suppose kani machri flows out from the cave during floods, how do you think it will survive?
- How do Kotumsar fishes differ from other fishes?

The fishes and other living organisms, are the biotic components of Kotumsar. The temperature, water, lime deposits, air, etc. are the abiotic components. The biotic and abiotic components in a habitat affect the survival of organisms living there.

17.2 The Interrelationship between Natural Components in a Habitat

Let us find out about the interrelationship of living and non-living components in a habitat.

Activity-1

Make a list of everything that you do as you wake up in the morning till you go to bed at night.

Make sure to include everything such as air you breathe, the toothbrush you use, milk or tea that you drink. Write the source from which you acquire these products. Like- the cotton clothes we wear are made of yarn that we acquire from the cotton plant. This is the source. Similarly, your plastic slippers are made of plastic which is obtained from the petrochemical industry.

Table 1

S.No.	Items	What it is made of?	Source
1.	Cotton kameez	Yarn	Cotton plant
2.	Slippers	Plastic	Petrochemical industry
3.			

The above table has been prepared based on items of daily use. If this is extended to a month or a year, it would become clearly evident that there are several things both living and nonliving in nature, which we depend upon.

- What are the nonliving components indispensable for us?
- What are the living components without which we cannot survive?

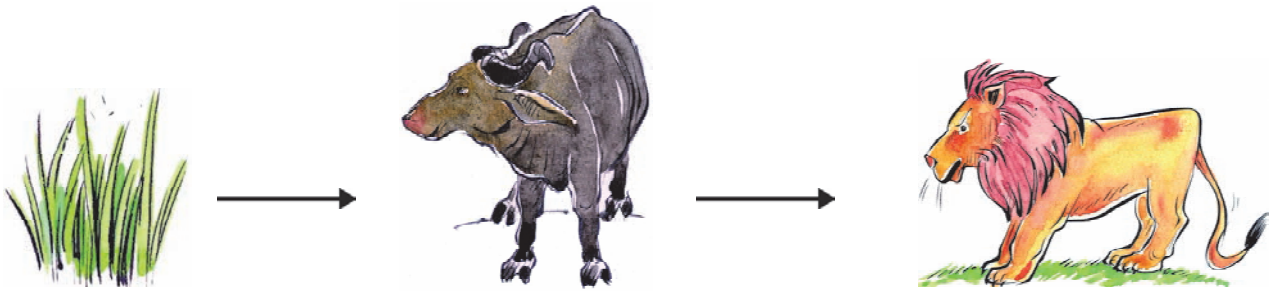
You may have now realized that we depend on many living organisms some more important while others less. Life of organisms depends on many others; thus, there is an interrelationship among organisms. Whether it is you or the rose plant, the lion or fungi all are dependent on other organisms. Each organism is part of a community and dependent upon several living and non-living components.

17.2.1 Interrelationship of Living Components

You have seen that organisms are dependent on each other due to either their reproductive, shelter or food related needs. Food and the energy acquired from them form the main basis of interrelationship of organisms.

Let us take an example to understand this -

The bison is found in the forests along rivers, marshy areas and wetlands of Chhattisgarh. It feeds on grasses and other plants. Food chain (energy flow) shows the dependence of the bison on grass. Food is prepared by grass by the process of photosynthesis. Thus, the flow of energy is from grass to bison (mind it, it cannot be reversed). If a lion feeds on bison, the relation can be shown as follows:



Grass Bison Lion

Figure number-3 : Food Chain

- Make a food web including several organisms along with grass, bison and lion.
- What would be the impact on this food web if the bison becomes extinct?
- Draw two food chains including yourself and state the interrelation between them.

Trophic Level

There are organisms in the food chain that are capable of producing food by the process of photosynthesis. These are called as "producers". There are others that depend on these producers (mainly plants) and other organisms for their food. These are the consumers, like the bison and the lion in this food chain. Here the bison is the primary consumer and the lion the secondary. . There can be other consumers in this chain as well, ending with organisms that would feed on dead and decaying matter, helping in their decomposition. These are the 'decomposers'. They break down complex organic matter to simpler ones which are required in nature.

The food chain may thus be written as -

Grass ⇨ Bison ⇨ Lion ⇨ Decomposers
(producers) (primary consumers) (secondary consumers)

*decomposers could be at any level of the food chain depending on what they decompose

Each level of the food chain here is known as a trophic level.

- Make a food chain including organisms of Kotumsar caves

Do you know?

During floods, the caves of Kotumsar are flooded with micro producers and consumers, plant parts, etc. Then the kanimachri feeds on these micro plants and animals. The fish of this cave otherwise feeds on the remains of dead organisms present here (detritus). As the food chain starts with dead and decaying matter rather than producers, it is called as detritus food chain.

Dead organisms and their remains → kanimachri → bat

17.2.2 Interrelationship between Biotic and Abiotic Components

The flow of energy and nutrients appears to be linear while observing a food chain. However, nutrients flow in nearly circular pathways (representing their reuse or conversion into other forms).

Here we come across the biotic components like grass, cow, man, microorganisms and abiotic components such as sunlight, fuel, soil, decomposed substances, etc. The arrows in the diagram denote the direction of nutrient flow as well as the relationship of the components.

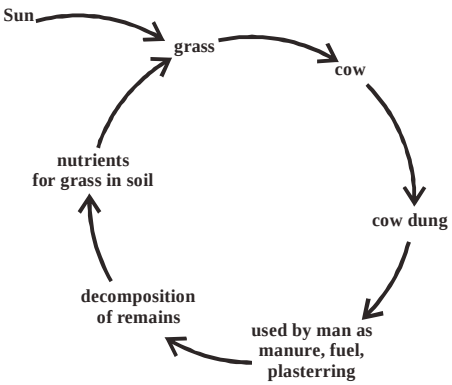


Fig. 4 : A cycle of nutrient flow

17.3 Diversity in Habitat

So far we have discussed the habitat of kani machri, let us now discuss the habitats of some other organisms.

Activity-2

You may have observed some plants that commonly grow in your area. The tendu (ebony), mahua, mango, jamun, babool, neem, ashok, etc. are some examples.

Choose any such plant and collect the following information regarding the same -

- Where did you see the plants of the species that you chose?

  (in the forest/in a field/on hills/near a river/far from a water body/ any other place)
- Write a description on the various elements such as the soil, temperature, water, etc. of the place where these plants grow..
- Write about some needs of the plant like how its pollens, seeds and fruits are dispersed etc. Your description can also include organisms that feed on the fruits of the plant.
- Could you now find out the habitat of the plant chosen by you?
- Try to draw a diagram of the plant.
- Make a list of some organisms from your surroundings that you see in larger numbers as compared to others.

Do you know?

Our intestine is a habitat for some micro organisms. We would not have been able to digest our food without their help. The removal of there micro organisms from our intestine would affect their survival and our health.

17.4 Changing Habitats in a Life Span

Now we shall discuss about an organism that inhabits different places during its life.

Brahmi duck is one such bird commonly known by other names like, Chakwa - Chakwi, Surkhaab, etc. It is not only found in India, but other Asian countries, Africa and Europe. In India, it is usually found during the summer months in the Himalayan regions of Laddakh, Sikkim and Arunachal Pradesh. This is where it breeds. During winter, the Brahmi duck leaves these colder regions of our country and other countries and is usually seen in the warmer regions of Central and Southern India. During this time the brahmi duck is also seen in Chhattisgarh. These birds thus change habitats with changing seasons.The

change of habitat in the lifespan of an organism as it travels from one habitat to another in a year is called as "migration". After a certain period of time, the organism returns to its previous habitat (often called as the "permanent habitat").



Fig. 5 : Brahmi Duck

The breeding grounds of Brahmi duck are usually near rivers, marshy areas, grasslands, fresh or saltwater lakes located up to around 5000meters above sea level. A secure shelter to lay eggs and more nutrition is required during the period of reproduction. The duck gets these favorable conditions in the Himalayan areas. Grass, cereals and other seeds, prawns, frogs, insects, etc. of this area are its food. As the food resources deplete with the onset of colder months, the Brahmi duck sets out to other areas rich in food resources. Group consists both of older and younger ducks.

- What is the habitat of a brahmi duck?
- Can brahmi duck be considered as a migratory organism? Why?

17.5 Adaptation to a Habitat

How are organisms adapted to their habitat?

We observe that organisms living and reproducing in a particular habitat have certain characters favorable for their survival. This is adaptation and there are several examples of it. For example fishes living in underground caves are often blind (just like the kani machri). A research done on such a fish, "the Mexican Tetra" has revealed several facts about the adaptation of the fish to its habitat. One of the facts is that the changes in salinity (kharepan) of the waters of the resource have an effect on the size of their eyes. The detailed account of the research has been given in Annexure to this chapter.



Fig. 6 : Mexican Tetra

We have studied so far that all living populations inhabit a habitat that fulfills its basic needs of nutrition, reproduction and shelter. An organism in a habitat is dependent on other organisms and also the non- living components there. A change in the environmental conditions would cause a change in the life style of the organism. Presently, we are changing our environment at a very fast pace which is not only affecting us but also the other organisms coexisting with us. Some of these changes are causes of serious problems faced by the earth.

Keywords

Habitat, migration, biotic component, abiotic component, interrelationship, food chain, food web, trophic level, producer, consumer, decomposer, scavenger



What we have learnt

- The basic needs of any organism are- food, shelter and reproduction.
- Habitat is that area for an organism that fulfills all its basic needs during its lifetime.
- There are several organisms that lead their life in different places and their habitats include different environments.

- There are some organisms that are migratory in nature, i.e., they spend some months of a year in an area and others in another.
- An organism adapted to a habitat will successfully survive and reproduce there.
- A food chain shows the relationship of an organism and its food, representing the flow of energy. It is a sequence of eating and being eaten that starts from producers and proceeds to the final consumers. The ultimate source of energy is the sun (producers as we know fix solar energy).
- Producers, consumers and decomposers are the different trophic levels of a food chain.
- Organisms also have a relationship with abiotic components of a habitat.

Exercise

1. Choose the correct option

   (i) We find kani machri, insects, snails, millipedes, remains of dead organisms and bats in Kotumsar caves. A correct food chain based on this information would be -

   (a) bats - millipede - kanimachri

   (b) remains of dead organisms - kanimachri - bats

   (c) bats - remains of dead organisms - kanimachri

   (d) insects - bats - snails

   (ii) Migratory organisms-

   (a) Lead their life in one place

   (b) Live in a new place every year

   (c) Live a few months in a year in one place and others in another place regularly.

   (d) They change their habitat once in their lifetime but do not return to their permanent habitat.

   (iii) We usually find the following at the first trophic level-

   (a) Producers (b) Primary consumers

   (c) Decomposers (d) Secondary consumers

2. Explain the biotic and abiotic components of the habitat of an organism of your choice.
3. Write an essay on: "what will happen if you reach such an island where there are no biotic components other than you?"

4. If humans were sent to the moon to live there - what all would have to be sent along with them? Make a list of all such things.
5. Make a food chain including a bear. Connect this with other food chains to form a food web.

Annexure

A group of American researchers carried out a study on a cave dwelling fish named "Mexican Tetra". How the fishes may have reached the cave? They speculated that during floods the ancestors of the fish may have entered the cave waters. This is how they may have been exposed to a new environment inside the cave.

First of all, the researchers tried to find out the difference in temperature, pH, dissolved oxygen, salinity, etc. of cave waters and that outside the cave. They found that salinity differed most. The cave waters were less saline as compared to that outside.

The researchers created cave water like conditions in the lab and grew embryos of the Mexican tetra which were not cave dwellers and found a lot of variation in the size of the eye of the fishes grown in cave like conditions. Some fishes had large eyes, while some had very small. There were several intermediate types as well. When the fishes with small eyes were breed, their young ones also had small eyes. It was not that variation did not exist before in the population that were not cave dwellers. It was just that the new environment of the cave may have increased it.